

Social Network Analysis and Visualization

QSS20: Modern Statistical Computing

What are social networks?

Agenda for today

- Social network analysis and visualization
- Notebook doing SNA

Network Analysis

- Social network analysis combines the **social science** and **network analysis**.
- Network structures are conceptualized with **graphs**.
- The study of how things are **connected**.

Many things should be thought of as networks

- Online interaction.
- Power grids.
- Genes interacting to produce you.
- Migration patterns.
- Game theory simulations.
- The **structure of social systems**



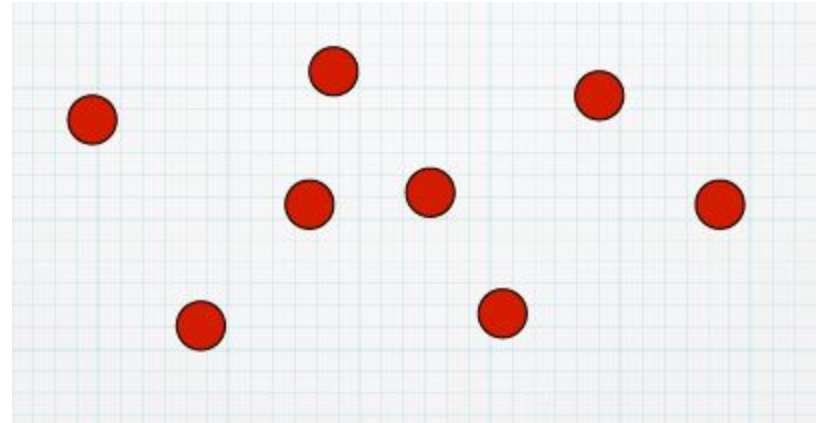
TikTok versus Instagram (or SnapChat; LinkedIn)

- What social media platforms do you use, if any?
- Which is more like a network?
- Which one do you feel more private on?
- Is there a sense of a network on anonymous social media (i.e. Fizz)?

What is a network?

Elements: Nodes and vertices

Could represent people, words, animals, etc.

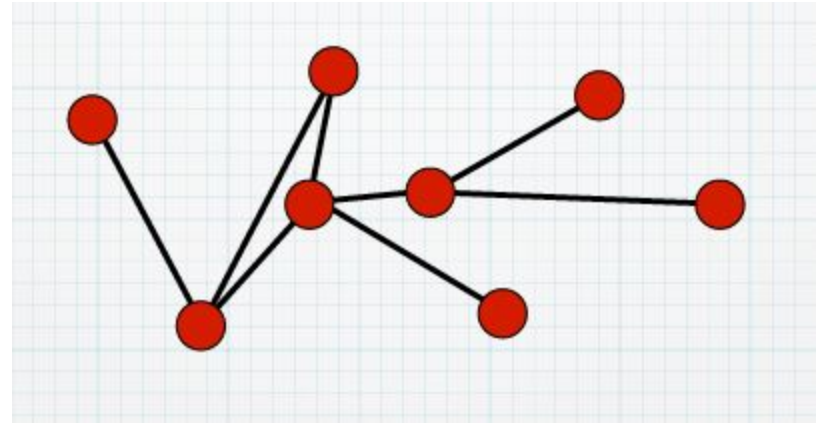


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Relationships: Edges



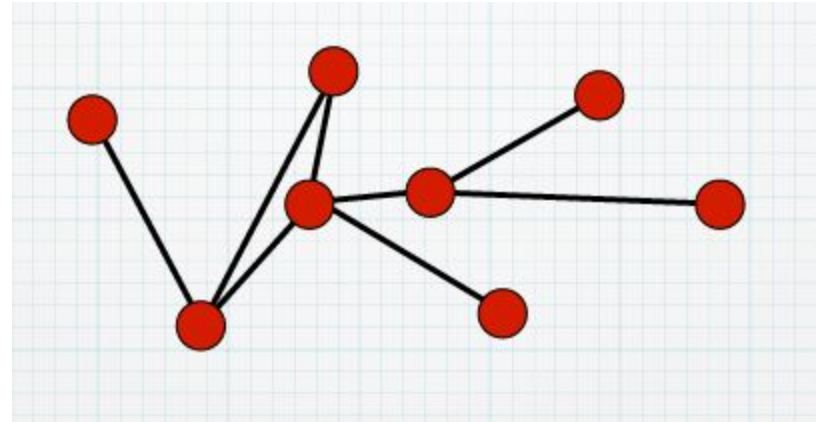
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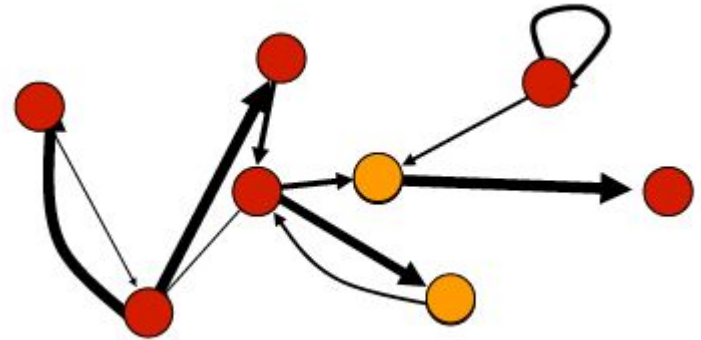
Relationships: Edges

Size of the network = # of nodes

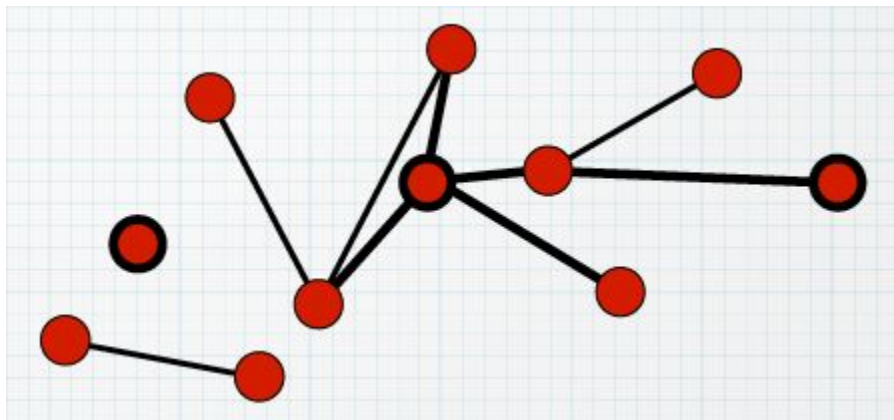


Types of Edges

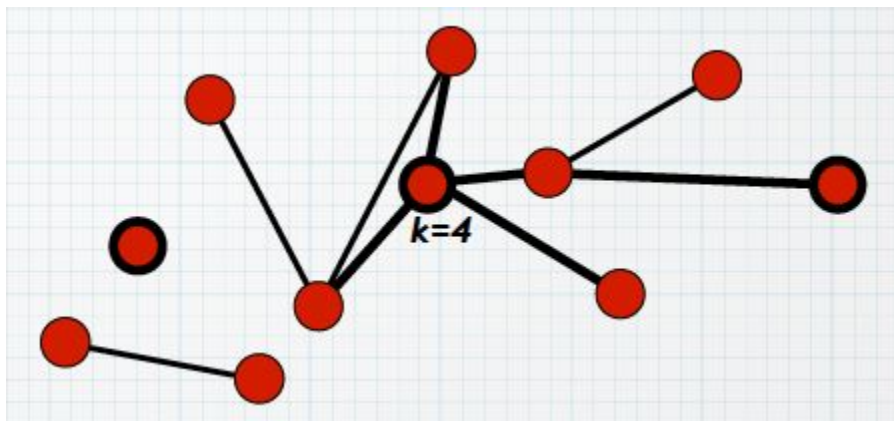
- **Directed Edges:**
I follow you but you don't follow me :(
- **Weights:**
The intensity of interactions between to nodes
- **In-degree:** Number of edges coming in
- **Out-degree:** Number of edges going out
- **Degree:** Sum of in and out degree
Can be seen as popularity!



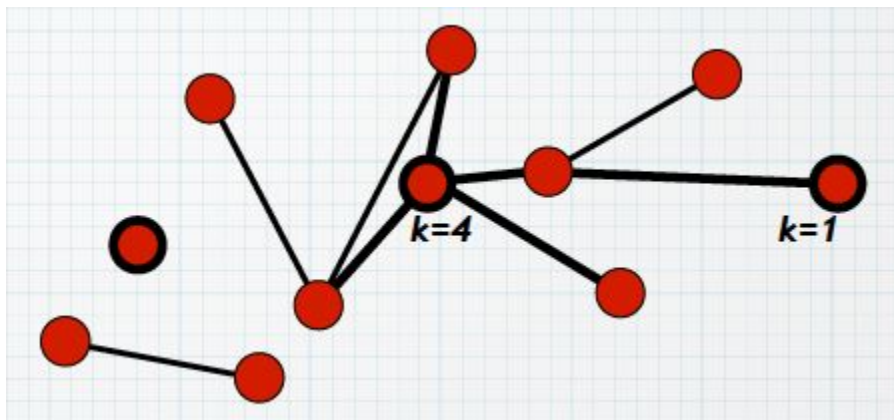
Degree



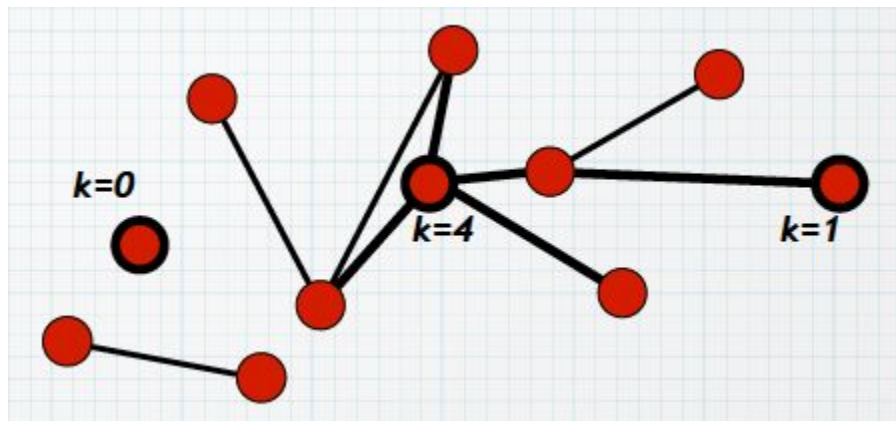
Degree



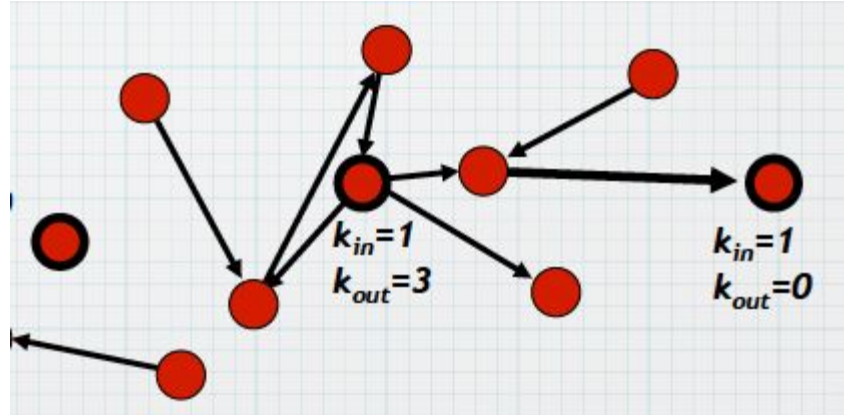
Degree



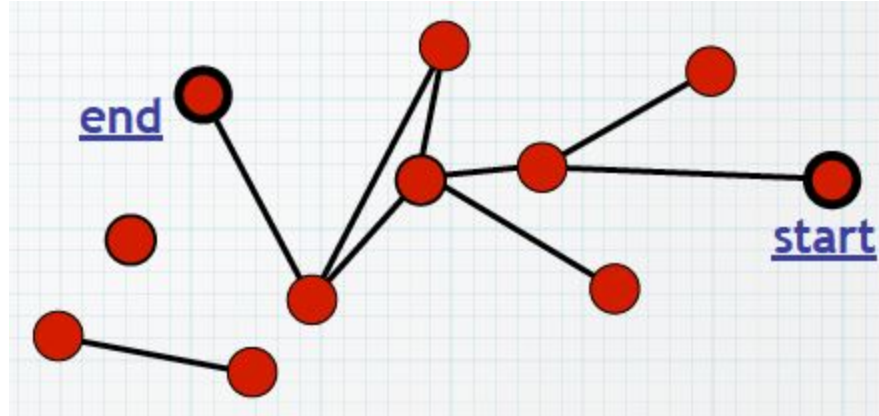
Degree



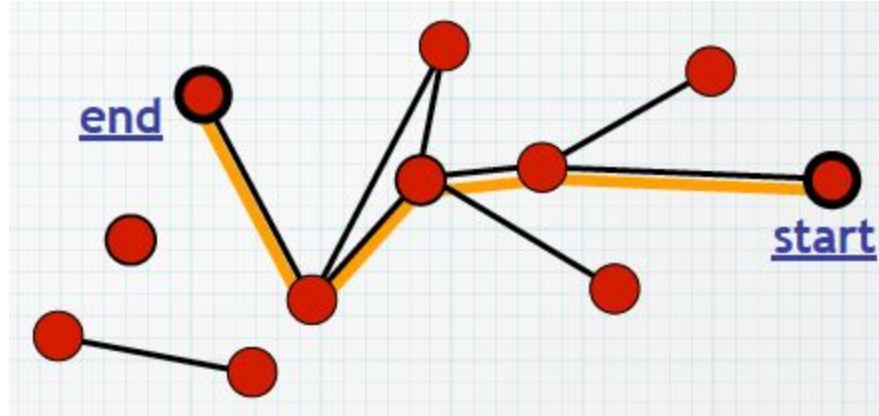
Directed Degree



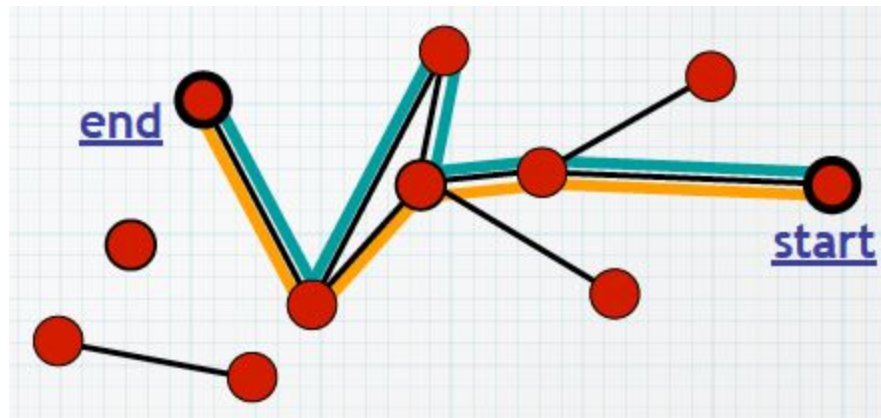
Distance



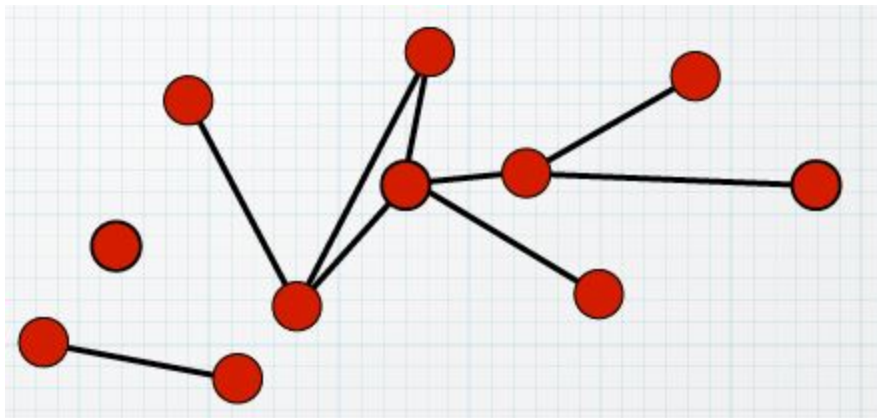
Distance



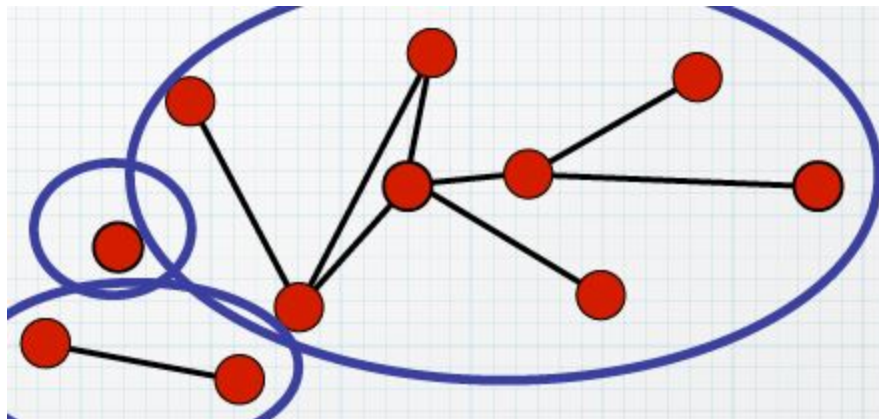
Distance



Giant Components



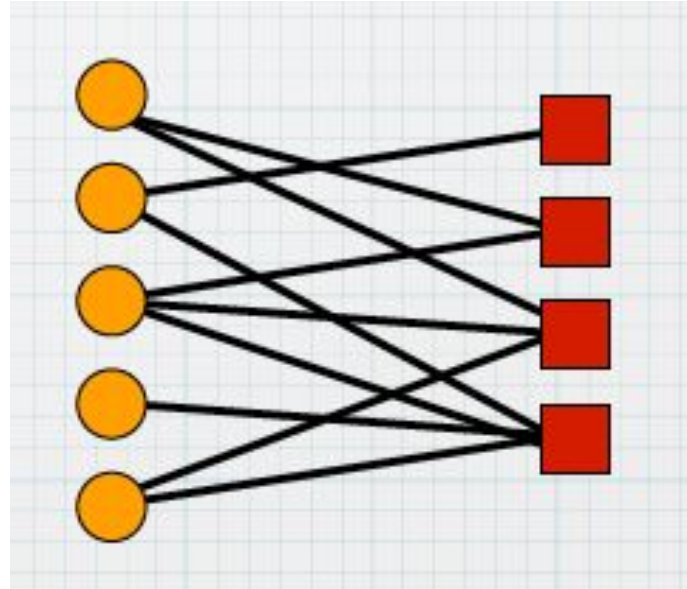
Giant Components



Bipartite Networks | Two classes of nodes

- Links between two classes of nodes
- Links ONLY happen between the two types

Does this look like a data structure we've seen?

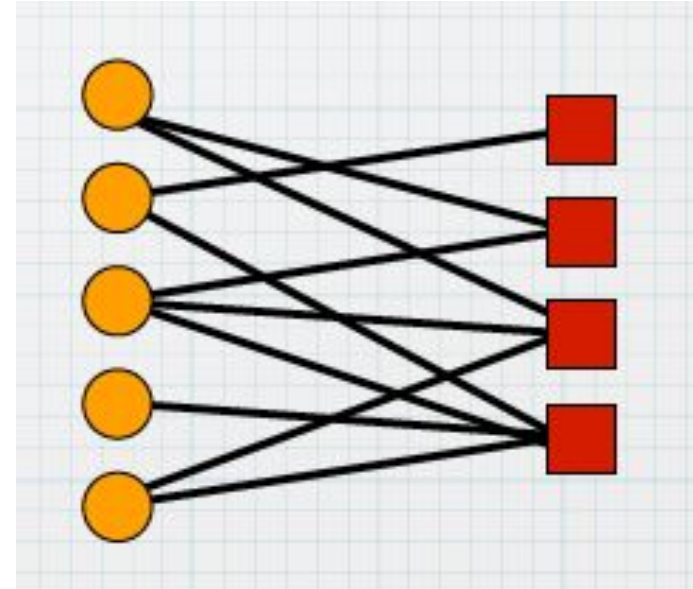


Bipartite Networks | Two classes of nodes

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Does this look like a data structure we've seen?

- Document to topic matrices!
- Gender to wage bracket
- F1 drivers to podium placements
- Race categories to sentence types



Summary of Definitions

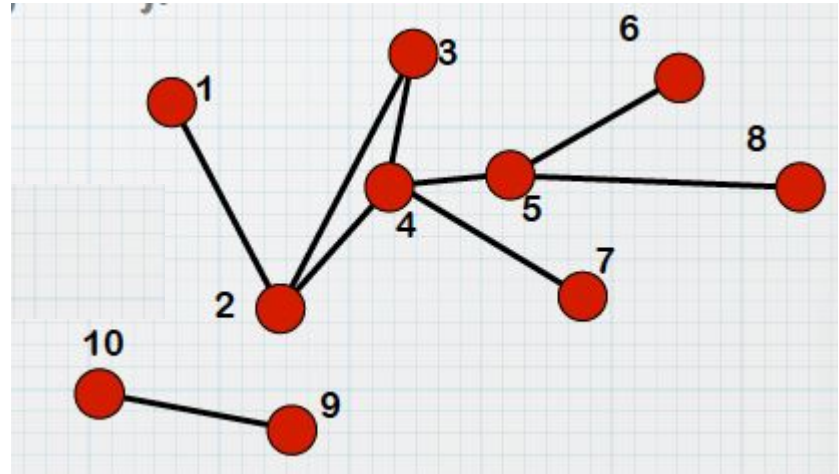
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(i.e. people and relationships)
- **Size** describes the total **number of nodes**
- **Edges** can be **directed** or **weighted**
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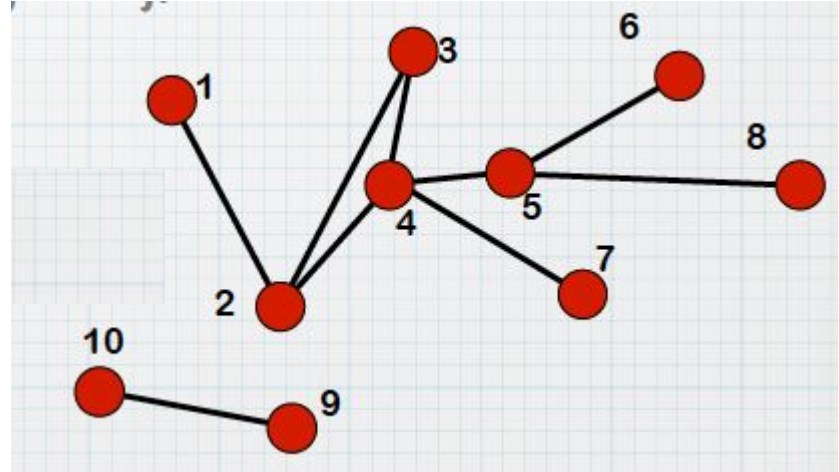
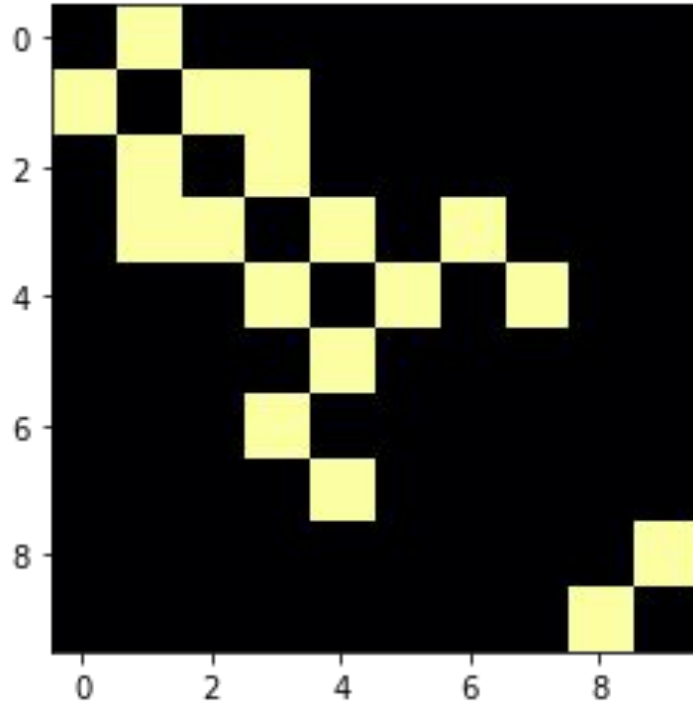
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- **Degree** describes the number of edges flowing in and out of a node.
 - In-degree is in; out-degree is out
- **Distance** is the number of hops between two nodes.
- The **giant component** is the biggest connected subgraph.
- **Bipartite graphs** represent links between two classes of nodes.

Representation of Graphs | Matrices (dataframes!)

	1	2	3	4	5	6	7	8	9	10
1	0	1	0	0	0	0	0	0	0	0
2	1	0	1	1	0	0	0	0	0	0
3	0	1	0	1	0	0	0	0	0	0
4	0	1	1	0	1	0	1	0	0	0
5	0	0	0	1	0	1	0	1	0	0
6	0	0	0	0	1	0	0	0	0	0
7	0	0	0	1	0	0	0	0	0	0
8	0	0	0	0	1	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	1
10	0	0	0	0	0	0	0	0	1	0

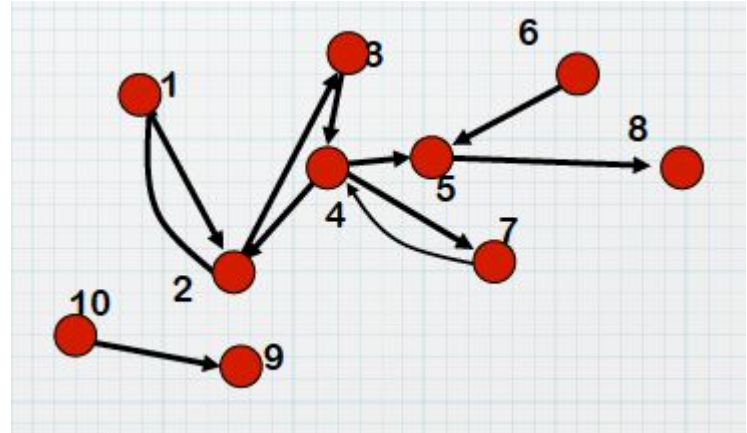


Representation of Graphs | Matrices (dataframes!)



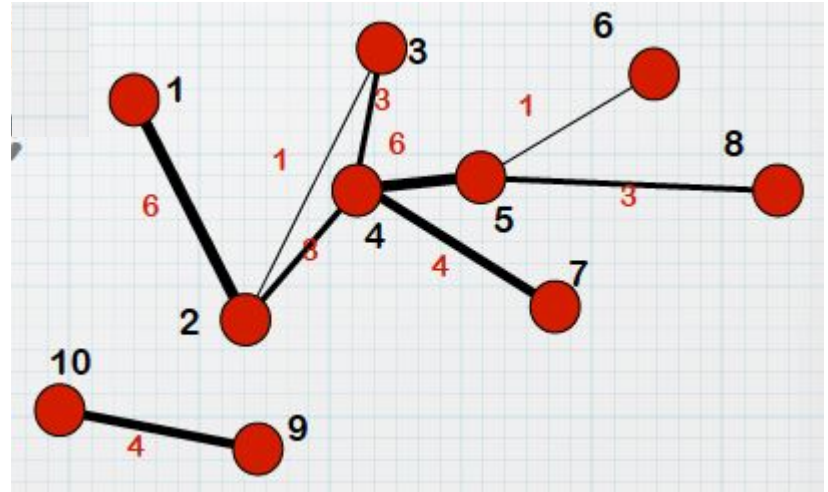
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4	0	1	0	0	1	0	1	0	0	0
5	0	0	0	0	0	0	0	1	0	0
6	0	0	0	0	1	0	0	0	0	0
7	0	0	0	1	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0
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10	0	0	0	0	0	0	0	0	1	0



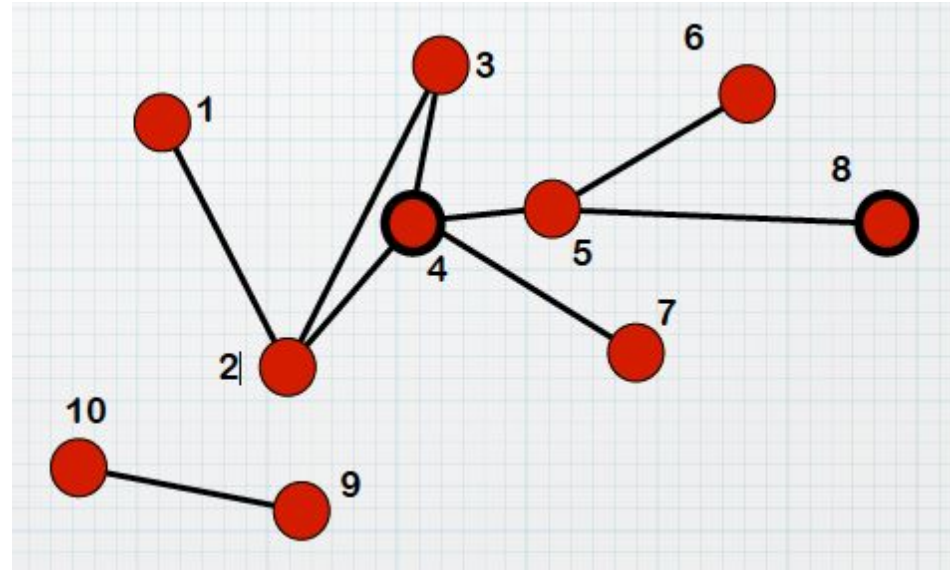
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1	0	6	0	0	0	0	0	0	0	0
2	6	0	1	3	0	0	0	0	0	0
3	0	1	0	3	0	0	0	0	0	0
4	0	3	3	0	6	0	4	0	0	0
5	0	0	0	6	0	1	0	3	0	0
6	0	0	0	0	1	0	0	0	0	0
7	0	0	0	4	0	0	0	0	0	0
8	0	0	0	0	3	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	4
10	0	0	0	0	0	0	0	0	4	0



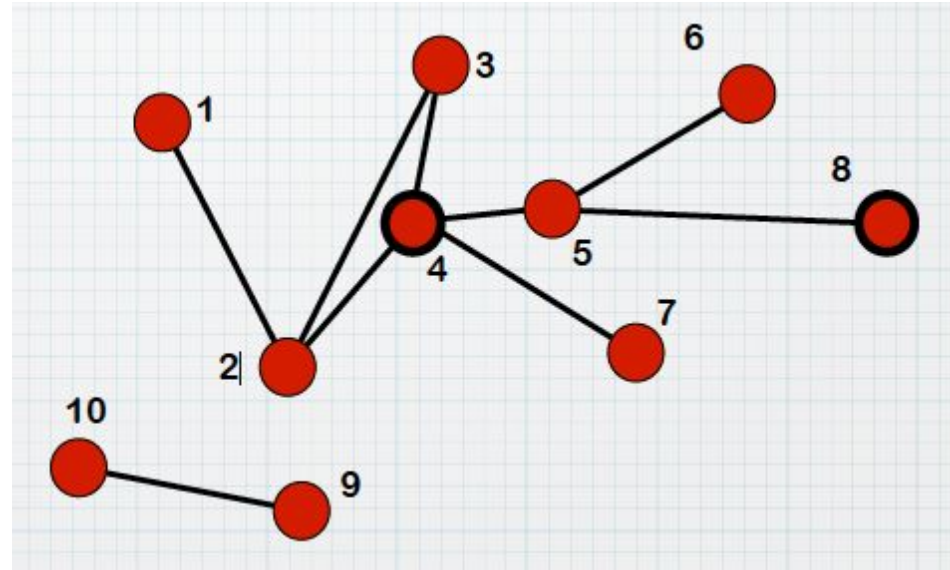
How can we find the degree with the matrix?

	1	2	3	4	5	6	7	8	9	10
1	0	1	0	0	0	0	0	0	0	0
2	1	0	1	1	0	0	0	0	0	0
3	0	1	0	1	0	0	0	0	0	0
4	0	1	1	0	1	0	1	0	0	0
5	0	0	0	1	0	1	0	1	0	0
6	0	0	0	0	1	0	0	0	0	0
7	0	0	0	1	0	0	0	0	0	0
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4	0	1	1	0	1	0	1	0	0	0
5	0	0	0	1	0	1	0	1	0	0
6	0	0	0	0	1	0	0	0	0	0
7	0	0	0	1	0	0	0	0	0	0
8	0	0	0	0	1	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	1
10	0	0	0	0	0	0	0	0	1	0

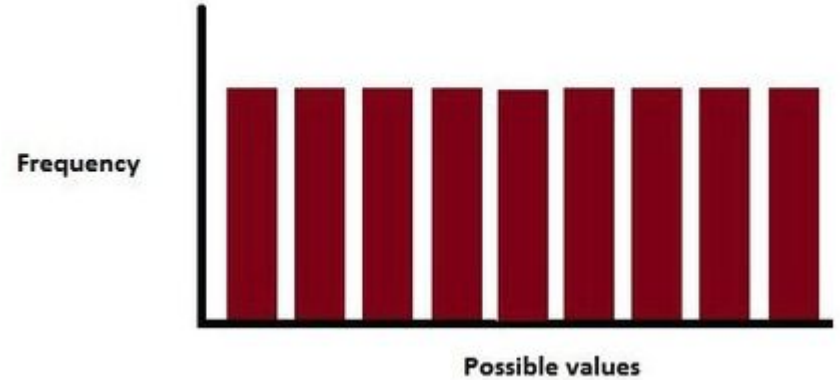
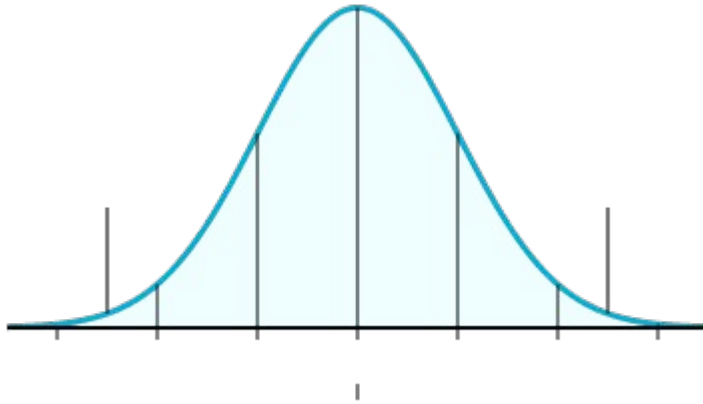


Social Network Analysis

The Study of How Things are **Connected**

What make something social?

Probability Distributions

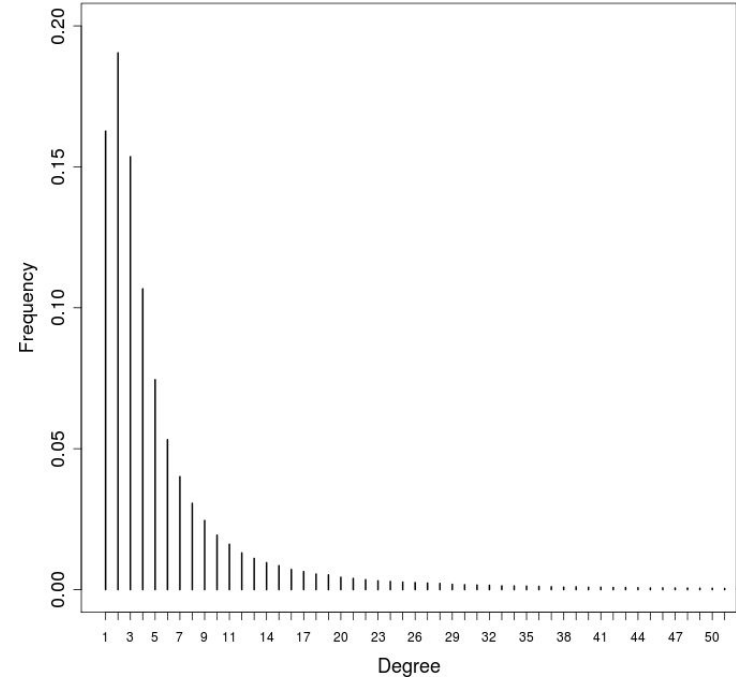
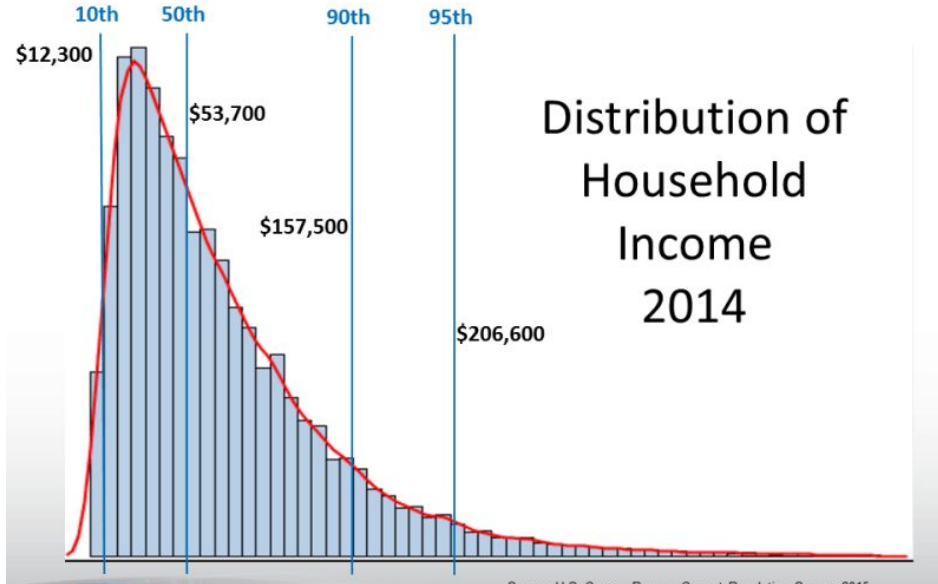


- Data exists in distributions.
- How do you know if a system is **social or not**?

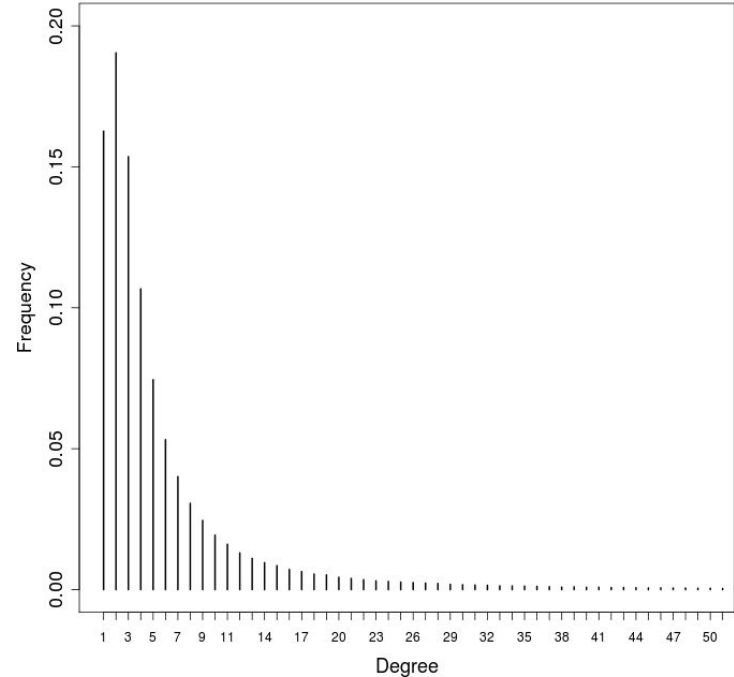
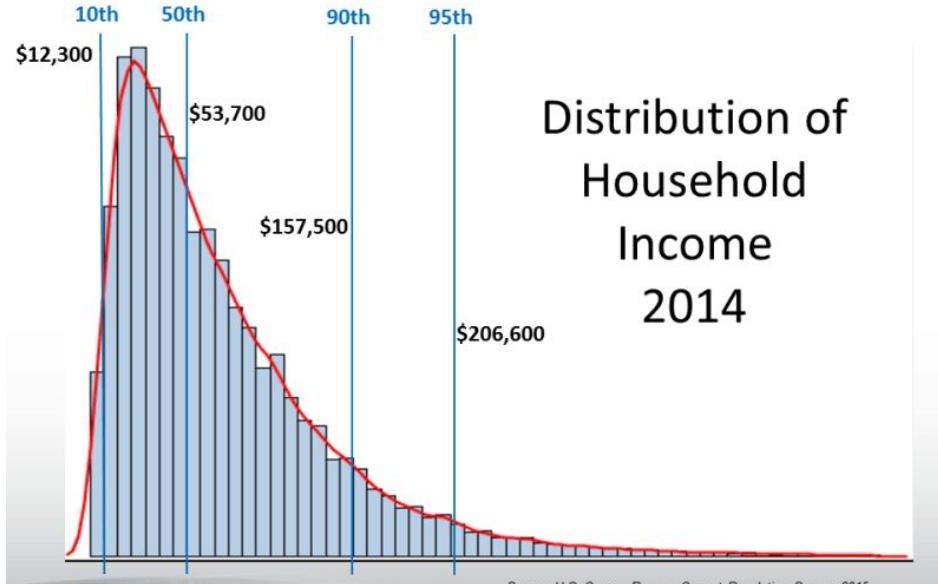
One mechanism, one distribution

- Humans are “**not natural**”
- Preferential attachment

What is common between wealth and popularity?

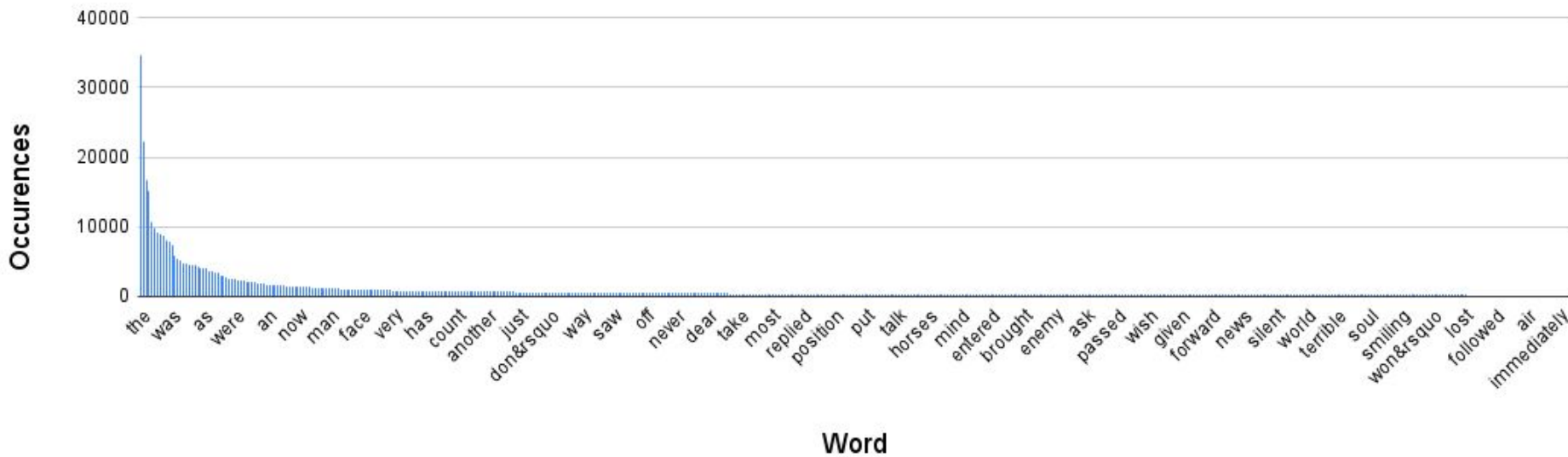


What is common between wealth and popularity?



Long tails

War and Peace: Word Occurences



Networkx

- Package to work with network objects
- Once you create your network object G, you can list...
 - Nodes: `G.nodes( )`
 - Edges `G.edges()`
 - Size of network: `len(G)`

Activity 1: Preprocess a network dataframe

- Load the raw Twitter data
- Filter the data to only have retweets and quotes
- Construct an edge list
- Create weights that are the average sentiment of all posts

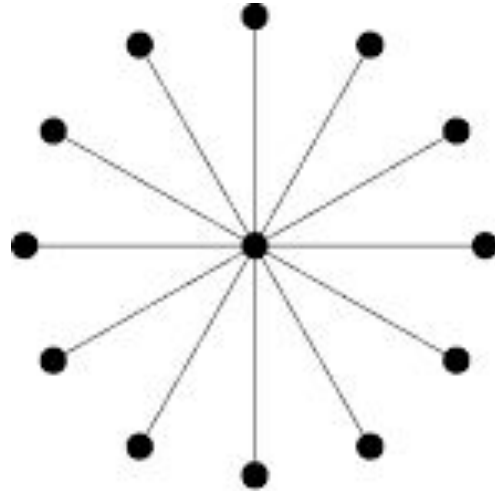
Notebook:

https://github.com/herbertfreeze/QSS20-X26/blob/main/activities/07_social_network_analysis.ipynb

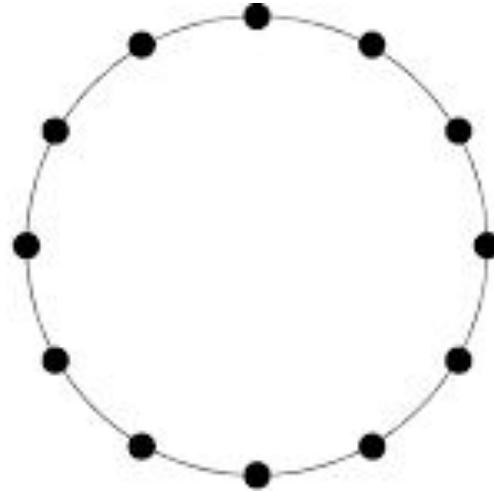
Who is Most Important?

What is centrality?

Centrality is a measure of how important a node is



A: Star Network



B: Regular Network

Popularity | Degree Centrality

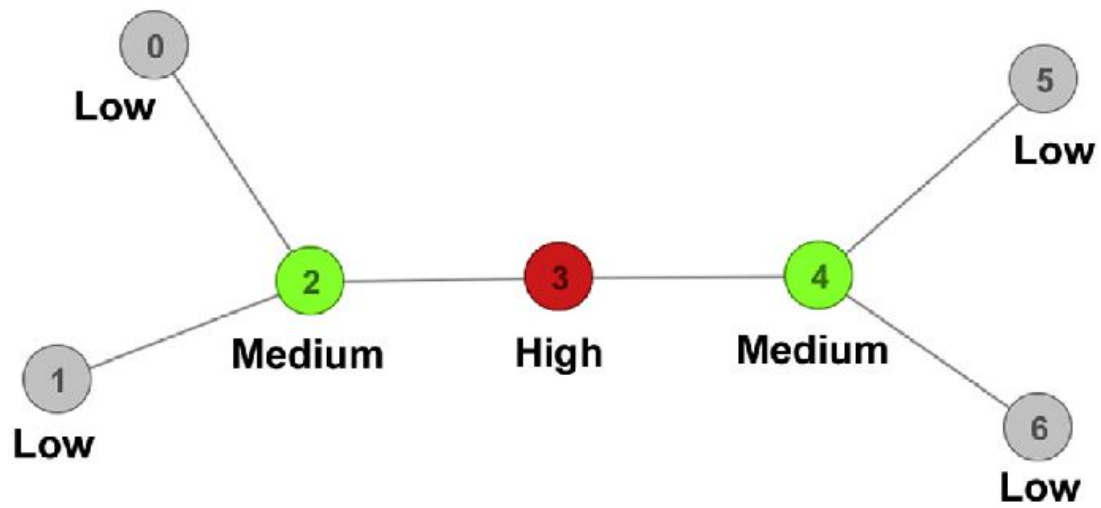
- **Degree centrality:** Total number of ties that a node has
- **In-degree centrality:** Total number of ties entering a node
- **Out-degree centrality:** Total number of ties entering a node
- How many steps does it take to calculate degree centrality for one node?

Brokerage | Betweenness Centrality

$$BC(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

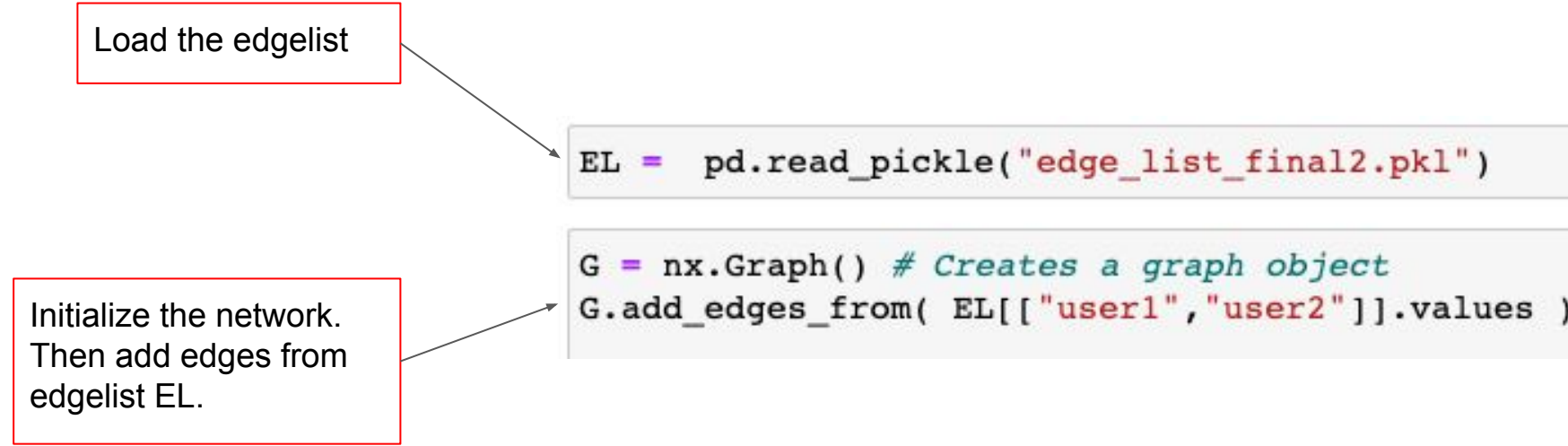
where:

- σ_{st} is the total number of shortest paths between nodes s and t .
- $\sigma_{st}(v)$ is the number of those shortest paths that pass through node v .
- The summation is taken over all pairs of nodes (s, t) in the graph, such that $s \neq t$ and $s \neq v$ and $t \neq v$.



Creating graphs with networkx

Load the edgelist

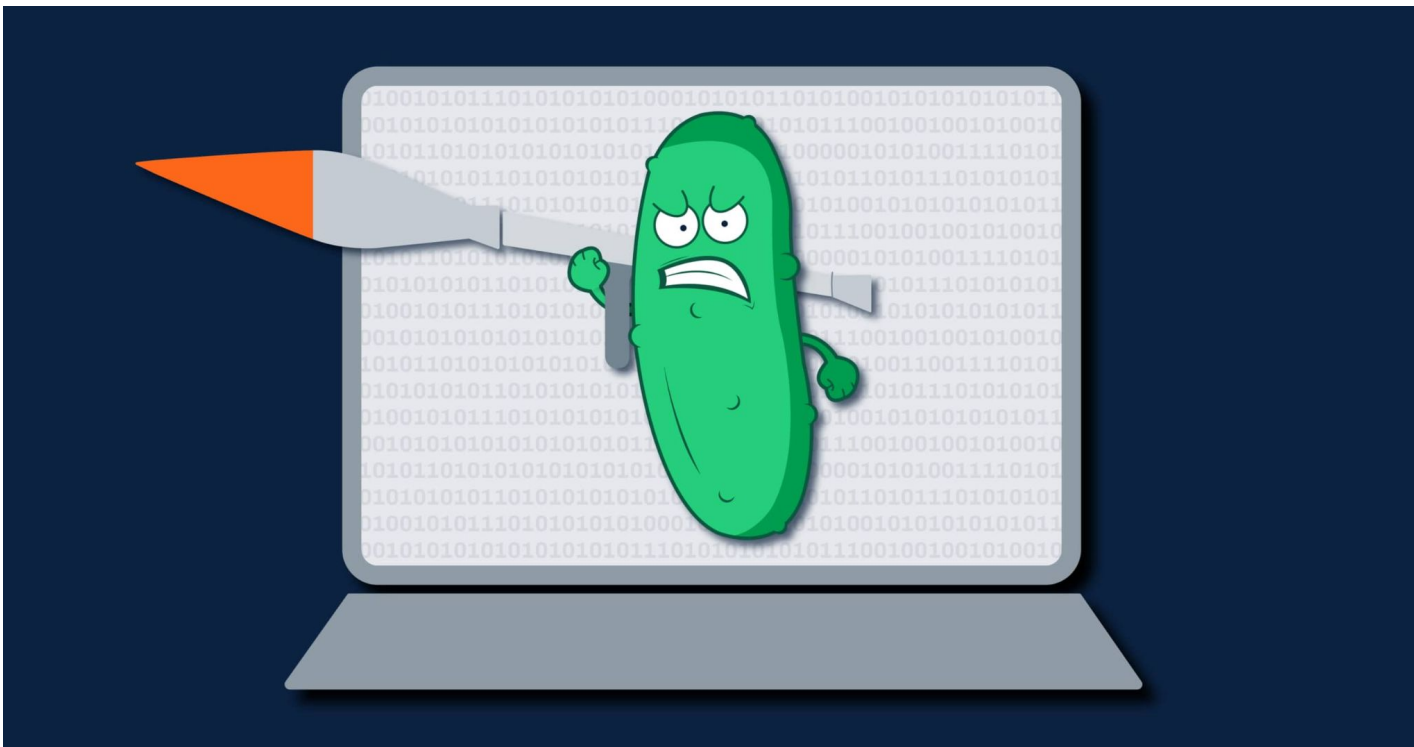


```
EL = pd.read_pickle("edge_list_final2.pkl")
```

Initialize the network.
Then add edges from
edgelist EL.

```
G = nx.Graph() # Creates a graph object  
G.add_edges_from( EL[["user1", "user2"]].values )
```

Why Pickle?



Creating graphs with networkx

For each of the nodes in GG, assign the node "group" with colors.



```
for k, v in GG.nodes(data=True):  
    v["group"] = id2color[k]
```

Activity 2

- Create a new column called degree in the user information table.
- Aggregate on the three types of users– democrats, republicans, and health officials. Who was the most popular (has the highest degree?).